

RESPONSE TO THE R40 REFEREE REPORT

QIAN QI
Peking University

This response addresses the report dated September 25, 2026 on *Certified Bellman Operators for Costly Policy Revision*. R43 builds on the published R42 manuscript rather than repeating or removing its results. The new work adds a regret-scaled global relaxation and a quadratic small-allowance theorem for controlled finite models, an observable-fiber convergence theorem for controlled atomic continuous states, and independently checked controlled computations. Inherited and newly executed evidence are distinguished, as are resolved arithmetic questions and remaining scientific questions.

1. SUMMARY OF THE SUBSTANTIVE REVISION

We agree that an exponential completeness theorem alone does not establish a practically useful numerical method. We also agree that exogenous aggregation cannot explain the controlled atomic maintenance evidence. R43 therefore adds two results that address the mathematical structure of controlled transitions directly.

First, relative to a strictly separated operating-optimal action, feasible departures are rare and implementation savings lie in a cone relative to operating regret. Eliminating the reference probability makes both factors in the remaining McCormick products proportional to the operating allowance. Reference-directed repair also changes only the rare nonreference probability mass. These facts give an explicit quadratic root-relaxation error bound as the allowance shrinks at fixed primitives. The proof identifies the action-gap and discount constants; it does not suppress them or claim a uniform polynomial algorithm.

Second, an observed covariate with invertible autonomous evolution labels finite controlled problems. The full Borel common-policy optimum disintegrates over these fibers. Validated fiber perturbations and complete finite solves yield convergent continuum certificates without smoothing atoms. The executed persistent-size economy has genuinely action-dependent condition transitions and a paired global continuum interval. It is not the original action-dependent continuous-condition model, and the article does not claim otherwise.

All 41 newly constructed endpoint objects pass independent reconstruction, covering 6,648 all-restart inequalities. All four continuum sums and four mutation rejections are retained. Both the eight controlled case specifications and source hashes were registered remotely before evaluation. A subsequent directed-arithmetic amendment is disclosed, with the original sources and partial results retained. The evidence is consequently described as predeclared cases with a disclosed amendment, not as an untouched code-frozen holdout. The full R42 article, supplement, response, computational results, and earlier history remain unchanged in the branch ancestry.

2. RESPONSES TO THE FOURTEEN SCIENTIFIC FINDINGS

F1: Theory and the continuous-state examples

New response. The main section “Controlled atomic models with observable covariates” proves exact disintegration and validated partition convergence for action-dependent regime kernels. The proof treats all Borel common policies, constructs measurable near-optimal families through the finite lattice and repair, and lifts policies using a label observable in the current state. It never substitutes total-variation smoothing for an atomic transition. The new controlled

continuum experiment executes this route with 17 independently checked finite fibers and four exact weighted sums. Its 16-cell interval is $[10.074215, 10.184476]$.

Inherited response and remaining boundary. R42 already executed the complete nonzero-error density-transfer budget and exact exogenous atomic aggregation. These remain intact. Neither the new observable-fiber assumption nor the inherited exogenous identities applies to the original action-dependent continuous-condition maintenance laws. The new theorem is a substantive controlled atomic extension, not a claim to have closed that particular model class.

F2: Severe repair constants and exponential completeness

New response. The savings-cone lemma and rare-departure repair yield a quantitative $O(\varepsilon^2)$ root width at fixed strict operating-action gap and fixed $\beta < 1$. This is an improvement over a tolerance-independent probability rectangle and a repair charge proportional only to the mixing fraction. The main theorem displays the full E_D , E_S , D_C , $\underline{d}$, and discount dependence. The supplement reports these constants even where they are numerically uninformative.

The three-orders-of-magnitude allowance study compares the same model and both relaxations. At $\varepsilon = 10^{-4}$ the scaled width is at most 0.000008657; the unscaled width is 0.009521686. No empirical slope is substituted for the theorem. Vanishing action gaps and high discount factors remain explicitly adverse, and generic completeness remains exponential. The inherited allowance ladder continues to handle finite undiscounted problems separately.

F3: The stronger McCormick comparator

New response. We use McCormick global programming as a component of the improved method, rather than attempting to defeat it with a weaker baseline. The new comparison holds the reference elimination, probability caps, operating-regret bounds, LP engine, tolerances, and candidate repair fixed. Only the savings cones and their product rectangles differ. The full table contains all 16 root computations, including 4 LP caps. There is a material interval reduction in 5 of the eight pairs. In the eight-regime, 32-period, four-action case the scaled interval has relative width 0.279%.

This establishes a useful structural tightening in a stated class, not blanket solver dominance. A stronger relaxation can be slower, and a time-capped scaled call can return a worse certificate. The old R40 McCormick comparison remains unchanged. The new work is an ablation of verified root relaxations, not a completed benchmark against every off-the-shelf branch-and-bound implementation. Construction, proposal, arithmetic, checking, dimensions, bytes, and cumulative high-water memory are reported with their actual meanings.

F4: All positive-cost primary randomized intervals remain open

Response. All 30 open primary intervals remain in the current article and are stated in the abstract. The 27 strict separations continue to mean randomized upper cost below deterministic lower cost, not randomized optimality. R42's exact horizon medians, maxima, and all 42 signed savings are retained. The new controlled eight-period small-allowance example supplies a distinct tightly bounded positive-cost randomized optimum. It is not substituted for, or counted as closing, an original primary row. Closing a substantial share of those original 30 rows remains a separate numerical target.

F5: One-shot bounds for the atomic primary model

New response. Observable-fiber disintegration and the partition proposition provide a genuine fixed-allowance convergence mechanism for controlled atomic continuum models. Finite solver

widths, primitive errors, and covariate partition errors have separate roles. The continuum experiment explicitly reports the surviving finite root widths, so it does not infer convergence from partition refinement alone.

Remaining boundary. The original affine-shock condition model has action-dependent continuous-state movements, rather than an independently recoverable covariate label. The new result does not prove convergence of its existing support/lottery construction. We retain that model, its full evidence, and its open width instead of replacing it with a claim about a different policy class.

F6: Nonlinear intervals and tight allowances

Inherited response. The complete finest $T = 32$, $N = 256$, $\varepsilon = 1/2$ nonlinear object received an independent arithmetic replay in R41; R42 preserved its exact report and hashes. R43 preserves and identifies that inherited check, rather than presenting it as a fresh execution. The wide nonlinear endpoints remain unchanged. The 0.05 exercises remain lower-bound-only because no matching feasible upper policy is certified. The new controlled continuum interval is paired at 0.01, but concerns a different economic state structure and is not used to erase this distinction.

F7: Size, actions, and breadth of the suite

New response. The controlled audit spans 3–32 regimes, 4–32 periods, three or four actions, two discount factors, and three allowances. All transitions are action dependent. These results complement, rather than duplicate, R42’s exogenous three-structure suite with up to 64 regimes, 64 periods, and five actions. Every registered controlled case remains in the table, including cases with no useful positive lower endpoint at the time cap. These are two designed economic families, not broad empirical validation or a multi-application external benchmark. The arithmetic amendment is visible in both the manuscript and protocol history.

F8: A difficult approximate operating oracle

Response. The inherited witness theorem and its complete margin accounting remain. R43’s new disadvantage coordinates use an exact finite operating solve, whose cost is included. We do not call that a difficult approximate operating-oracle experiment. Executing the witness-driven theorem where operating approximation is itself the dominant difficulty remains uncompleted. The response addresses the computational method with a new finite structural rate and a controlled continuum reduction, not by mislabeling the available oracle.

F9: Restrictive transfer assumptions and hidden constants

New response. The fiber route uses a finite matrix norm while retaining the exact atomic covariate coordinate. It does not require a reference-measure decomposition of a smoothed covariate transition. The initial law is specified as an exact independent regime/covariate mixture. Its generic perturbation proposition displays all primitive errors; the executed scale model additionally admits exact monotone endpoint bounds with rational weights. The table separates finite endpoint width from the remaining partition contribution. R42’s density budget, including its failed tightening condition and explicit repair constant, is retained unchanged. These results do not claim dimension-free continuous-state scalability or estimated model-error radii.

F10: Economic discipline and calibration

Response. The revised controlled examples make the economic channel explicit: interventions alter future condition probabilities, while observed establishment size changes the operating-loss allowance in base units and changes implementation costs. All normalization and initial laws are stated. They remain designed known-model economies. No estimated transition law, monetary welfare estimate, sampling uncertainty set, or empirical prevalence claim is fabricated. The new methodological contribution is carried by the controlled small-allowance theorem and observable-fiber convergence result, not by claiming an empirical application that has not been conducted. The inherited administration and policy-uniform perturbation propositions remain with their assumptions.

F11: Hardest-object verification and trust boundaries

New response. Every new controlled object, including every 32-period and 32-regime time-capped object, is independently reconstructed. The reader checks actual deployed policies against every original restart constraint, as well as every rational relaxation lower endpoint. A separate audit verifies the declared primitive generator and all continuum sums. No adjacent-object verification label is transferred to an unchecked row. The supplemental coverage map distinguishes these checks from analytic theorem proofs, source-hash identity, and publication integrity. The inherited finest nonlinear replay remains separately labeled.

F12: Novelty relative to existing methods

New response. The paper now states a theorem-level distinction rather than relying on repository engineering. McCormick envelopes, bound tightening, and linear programming are not new. What is established here is a Bellman disadvantage/savings representation whose two product factors contract with the operating allowance, together with rare-departure common-policy repair that preserves a quadratic root-width bound. The controlled observable-fiber theorem further gives an exact optimization identity over all Borel common policies and a constructive approximation mechanism without total-variation smoothing. The existing occupation-measure compatibility discussion is preserved: dropping common-policy compatibility still changes the target.

These are conditional guarantees. They do not establish universal computational superiority, novelty of classical LP ingredients, or convergence of the original unrestricted action-dependent continuous-condition model.

F13: Marginal contribution of the local LP

Response. The connected local-support construction and all its original accounting remain, and its local-relaxation target is unchanged. R43 adds a different same-object finite relaxation ablation rather than attributing that new work to the earlier local LP. A matched marginal local-work/global-node experiment on the original continuum model has not been completed. We do not infer its marginal economic value from a different controlled finite experiment. The detailed local component remains supplementary evidence, without deletion of its derivation or failed runs.

F14: Finite horizon, zero tolerance, and discounting

Inherited and new response. R42 extended finite-horizon repair to $\beta = 1$ with an explicit allowance ladder, provided an exact zero-tolerance operating-action restriction, and stated an

infinite-horizon discounted tail enclosure with its exact-terminal-value trust boundary. All are retained. The new quadratic root theorem deliberately states the stronger conditions it needs: fixed $\beta < 1$ and a strictly positive operating-action gap. It is not used to claim a uniform limit at $\beta = 1$ or action ties. The stopped continuous-control program remains archival and is not relabeled solved.

3. DISPOSITION OF THE ADDITIONAL PRESENTATION COMMENTS

Comments 1–5 are addressed by the abstract’s explicit original-model boundary and 30 open intervals, and the retained exact median/maximum and signed deterministic-savings tables. The direction of every strict lottery comparison remains upper-versus-lower.

Comments 6–10 are addressed in the new tables by reporting $nT(m-1)$, all three/four-action configurations, one-root-node status, separate work categories, and the exact scope of the 30-second LP cap. The original 60-second iterative cap remains described separately. A numerical worst-case full-tree count for every original R40 row is not newly supplied; the new structural theorem instead reports its own explicit root-width constants. No root ablation is misrepresented as a multi-node complexity experiment.

Comments 11–14 retain the witness-budget and density-error tables and the distinction between action-distribution total variation and the full operator convention. The new fiber matrix norm is explicitly a finite-coordinate ℓ_1 bound, not continuum total variation. The initial regime law and covariate distribution are exact specified primitives. The action-gap and discount constants of the new rate are tabulated rather than hidden inside asymptotic notation.

Comments 15–18 retain nonlinear feasibility, threshold-crossing, and coverage labels; the finest nonlinear independent result is explicitly inherited. Tighter-allowance lower-only rows stay lower-only. Development, predeclared designed experiments, and empirical validation are not combined into a success count.

Comments 19–20 retain the economic-overhead and policy-uniform model-error assumptions, without inventing estimated uncertainty radii. State/action-dependent overhead would have to enter the actual cost and, if it changes dynamics or rewards, the corresponding operating constraints. No such unexecuted modification is incorporated silently.

Comments 21–23 are addressed by separate constructor/checker timings, byte counts, human-readable source-hash equality, and retained explicit common-policy occupation compatibility. The new primitive-contract audit imports no constructor. Analytic disintegration and rate proofs are not described as machine-checked.

Comments 24–25 are addressed by preserving the stopped-control analysis and full previous papers in the archive, while the current main article is self-contained. No inherited scientific file is deleted or overwritten. A branch-level preservation audit and current review index identify the exact current documents.

4. EVIDENCE ACCOMPANYING THIS RESPONSE

The current article, supplement, response, and computation report are `ECTA_R43.pdf`, `SUPP_R43.pdf`, `RESPONSE_R43.pdf`, and `COMPUTATION_R43.pdf`. The replication directory contains the complete registered and amended protocols, readable constructor and independent checker sources, compressed rational proof objects, all results, mutation outcomes, and publication manifests. The published R42 source and results remain available unchanged, and no other paper or review branch is modified.